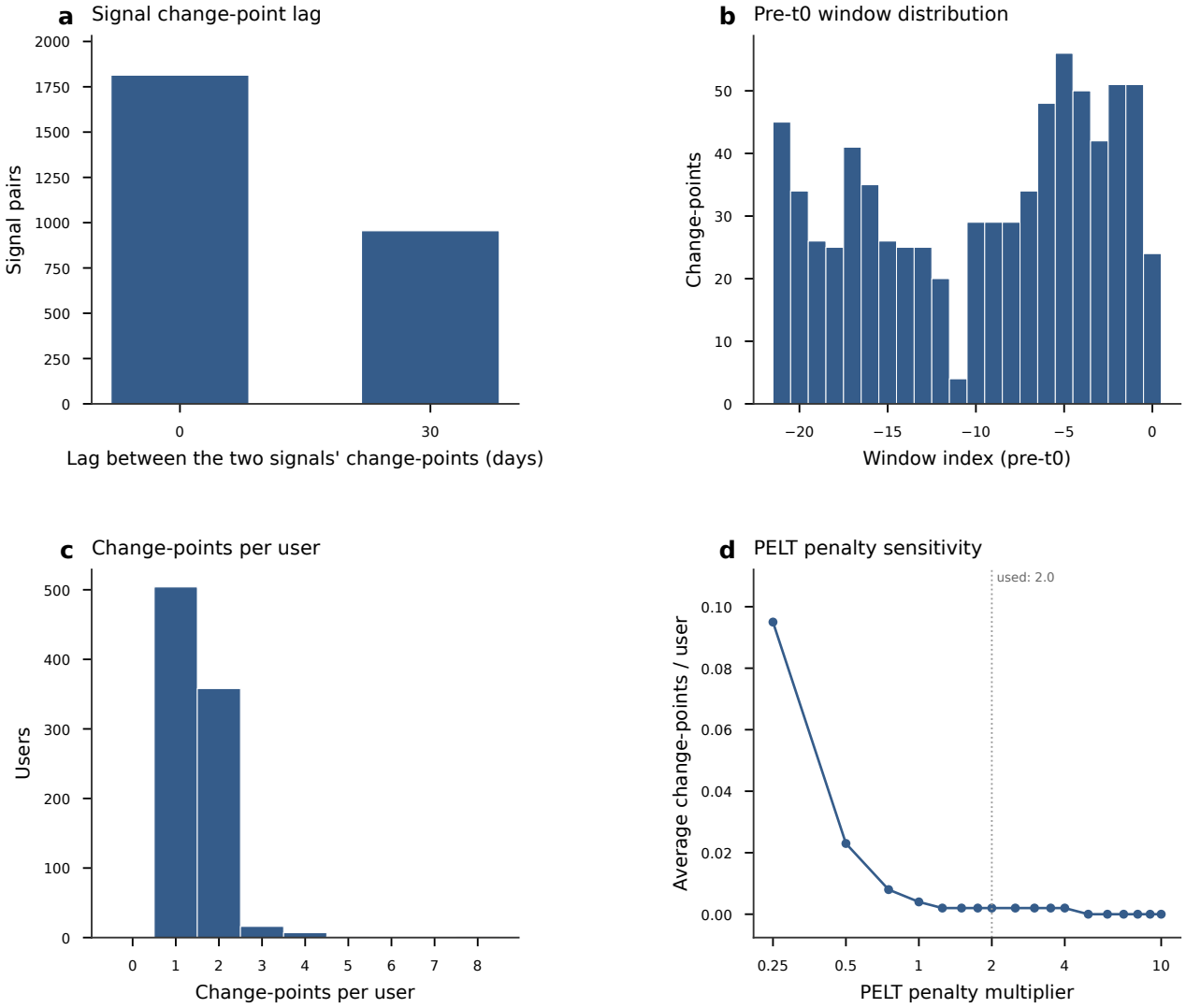
a
Signal change-point lag
Signal pairs
2000
1750
1500
1250
1000
750
500
250
0
0
30
Lag between the two signals' change-points (days)
b
Pre-t0 window distribution
Change-points
50
40
30
20
10
0
−20
−15
−10
−5
0
Window index (pre-t0)
c
Change-points per user
Users
500
400
300
200
100
0
0
1
2
3
4
5
6
7
8
Change-points per user
d
PELT penalty sensitivity
used: 2.0
Average change-points / user
0.10
0.08
0.06
0.04
0.02
0.00
0.25
0.5
1
2
4
10
PELT penalty multiplier